

Wi-Fi Training Program

Module 4 – Assignment Questions & Answers

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1. What is the significance of MAC layer and in which position it is placed in the OSI model?

Answer:

The MAC (Medium Access Control) layer is one of the most critical layers in wireless networking. It sits at Layer 2 (Data Link Layer) of the OSI model, specifically as the lower sublayer of the Data Link Layer. The upper sublayer of Layer 2 is the LLC (Logical Link Control).

Position in OSI Model:

OSI Layer	Layer Number	Sublayer
Network	Layer 3	IP, routing
Data Link	Layer 2 (Upper)	LLC – Logical Link Control
Data Link	Layer 2 (Lower)	MAC – Medium Access Control ← Here
Physical	Layer 1	PHY – Radio signals, OFDM, modulation

Significance of the MAC Layer:

- Channel Access Control: Determines when and how devices access the shared wireless medium using CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).
- Frame Formatting: Defines the structure of 802.11 frames including addressing, control fields, and data.
- Addressing: Uses MAC addresses (48-bit) to identify source and destination devices on the network.
- Reliability: Handles acknowledgment (ACK) of frames and retransmission of lost frames.
- Association and Authentication: Manages the process of clients joining the BSS (Basic Service Set).
- Power Management: Controls when client devices can enter sleep mode and when the AP buffers frames for them.
- Security: Implements encryption (WPA2/WPA3) and authentication (802.1X/EAPOL) at the MAC level.
- QoS: Prioritizes traffic using 802.11e (WMM) Access Categories for voice, video, best-effort, and background traffic.
- Fragmentation and Aggregation: Splits large frames into smaller fragments or combines small frames for efficiency.

In short, the MAC layer is the intelligence of the wireless network — it decides who transmits, when they transmit, how frames are formatted, and how reliability and security are maintained.

2. Describe the frame format of the 802.11 MAC header and explain the purpose of each field.

Answer:

An 802.11 MAC frame consists of a MAC Header, Frame Body (payload), and FCS (Frame Check Sequence). The MAC header contains multiple fields that control how the frame is processed.

Field	Size	Purpose
Frame Control	2 bytes	Defines the frame type (Management/Control/Data), subtype, and various flags (To DS, From DS, More Frag, Retry, Power Mgmt, More Data, Protected, Order)
Duration/ID	2 bytes	Specifies the time (in microseconds) the channel will be occupied — used by other devices to update their NAV (Network Allocation Vector) for virtual carrier sensing
Address 1 (RA)	6 bytes	Receiver Address — the immediate recipient of the frame (next hop). Always present.
Address 2 (TA)	6 bytes	Transmitter Address — the device that is transmitting the frame. Always present.
Address 3	6 bytes	Depends on To DS / From DS bits. Could be BSSID, source, or destination address depending on frame direction.
Sequence Control	2 bytes	Contains Fragment Number (4 bits) and Sequence Number (12 bits) — used to reassemble fragmented frames and discard duplicates
Address 4	6 bytes	Used only in WDS (Wireless Distribution System) or mesh — carries the original source address in 4-address frames
QoS Control	2 bytes	Present in QoS Data frames — specifies the Traffic Identifier (TID) / Access Category for prioritization
HT Control	4 bytes	Present in HT/VHT frames — carries link adaptation and beamforming feedback information
Frame Body	0–7951 bytes	The actual payload — management info, control data, or upper-layer data (e.g., IP packets)
FCS	4 bytes	Frame Check Sequence — CRC-32 checksum to detect transmission errors

Frame Control Field – Key Flags:

- To DS / From DS: Indicates the direction of frame flow (0/0 = IBSS, 0/1 = from AP to client, 1/0 = from client to AP, 1/1 = WDS)
- Retry: Set to 1 if the frame is a retransmission
- Power Management: Indicates if the sender is entering power save mode after this frame
- Protected Frame: Set to 1 if the frame body is encrypted (WPA2/WPA3)

3. Please list all the MAC layer functionalities in all Management, Control and Data plane.

Answer:

Management Plane – Frame Types and Functions:

Management frames handle the establishment and maintenance of communication between devices. They are never encrypted in legacy 802.11 (PMF – Protected Management Frames in 802.11w adds protection).

Frame Subtype	Function
Beacon	AP periodically broadcasts its presence, SSID, supported rates, and capabilities
Probe Request	Client broadcasts to find available APs (active scanning)
Probe Response	AP responds to a Probe Request with its capabilities
Authentication	First step of joining — Open System or SAE authentication exchange
Association Request	Client requests to join the BSS and sends its capabilities
Association Response	AP accepts or rejects the association request
Reassociation Request/Response	Used during roaming to move association from one AP to another
Disassociation	Terminates the association — can be sent by AP or client
Deauthentication	Terminates the authentication — forces client to restart join process
Action	Carries specific management actions (spectrum management, QoS, block ACK setup, beamforming)

Control Plane – Frame Types and Functions:

Control frames assist in the delivery of data and management frames. They ensure reliable delivery and manage channel access.

Frame Subtype	Function
ACK (Acknowledgment)	Confirms successful receipt of a unicast frame — sent by receiver after every unicast data/management frame
RTS (Request to Send)	Sent before a large data frame to reserve the channel and avoid hidden node collisions
CTS (Clear to Send)	Sent in response to RTS — confirms channel is clear; also used as CTS-to-self for protection
Block ACK Request (BAR)	Requests a block acknowledgment for a set of frames sent in an aggregated burst
Block ACK (BA)	Acknowledges multiple frames at once using a bitmap — one ACK for up to 64 frames
PS-Poll	Sent by a client in power save mode to request buffered frames from the AP
CF-End	Signals the end of a contention-free period (PCF mode)
Trigger Frame	Used in 802.11ax (Wi-Fi 6) to schedule uplink OFDMA/MU-MIMO transmissions from clients

Data Plane – Frame Types and Functions:

Data frames carry the actual user payload (IP packets, etc.) between devices.

Frame Subtype	Function
Data	Basic data frame carrying upper-layer payload
Null Data	Carries no payload — used by clients to notify AP of power save state change (no data, just Power Management bit)
QoS Data	Data frame with QoS Control field — carries Traffic Identifier for WMM prioritization
QoS Null	QoS version of Null frame — used for power management signaling with QoS-capable APs
Data + CF-Ack	Data frame with piggybacked ACK for the previous frame — used in PCF mode

4. Explain the scanning process and its types in detail.

Answer:

Scanning is the process by which a wireless client discovers available Access Points (APs) in its vicinity. It is the first step before a client can associate to a network. There are two types of scanning: Passive and Active.

1. Passive Scanning:

In passive scanning, the client does not transmit anything. It simply listens on each channel for Beacon frames that APs periodically broadcast.

- APs broadcast Beacon frames approximately every 102.4 ms (default Target Beacon Transmission Time – TBTT).
- The client switches to each channel, waits for a beacon (or a configured dwell time), records AP information, then moves to the next channel.
- Information gathered from beacons: SSID, BSSID, supported rates, channel, security capabilities (RSN IE), and QoS info.
- Advantage: Client remains silent — useful in sensitive environments and saves some transmit power.
- Disadvantage: Slower than active scanning — must wait for beacon on each channel (up to 102.4 ms per channel).
- Used By: Devices that want to minimize transmission (e.g., IoT sensors) or in regulatory domains that require passive scanning on certain channels (e.g., DFS channels in 5 GHz).

2. Active Scanning:

In active scanning, the client transmits a Probe Request frame on each channel and waits for Probe Response frames from APs.

- The client sends a Probe Request — either with a specific SSID (directed probe) or with a broadcast SSID (wildcard probe for all APs).
- All APs on that channel that match the SSID respond with a Probe Response containing their capabilities.
- The client collects all responses, then moves to the next channel.
- Advantage: Much faster than passive scanning — does not need to wait for a beacon.
- Disadvantage: Generates traffic on the medium; may reveal the client's preferred SSIDs (privacy concern).
- MinChannelTime and MaxChannelTime: The client waits at least MinChannelTime for a response. If it hears activity, it waits up to MaxChannelTime before moving on.

Scanning During Roaming:

When a connected client needs to roam, it performs background scanning — scanning on off-channels while still connected to the current AP — to find better APs before signal quality degrades. Technologies like 802.11k (neighbor reports) and 802.11r (fast BSS transition) assist this process.

5. Brief about the client association process.

Answer:

The client association process is the series of steps a wireless device must complete before it can send and receive data on a Wi-Fi network. The process involves three phases: Scanning, Authentication, and Association.

Step 1 – Scanning (AP Discovery):

- The client performs passive or active scanning to discover available APs.
- It collects AP information (SSID, BSSID, RSSI, security type, capabilities).
- The client selects the best AP based on signal strength and capabilities.

Step 2 – Authentication (802.11 Open Authentication):

- The client sends an Authentication Request frame to the selected AP.
- The AP responds with an Authentication Response (Status: Success).
- Note: This 802.11 authentication is not security authentication — it is just a handshake to establish the 802.11 link. Real security authentication (WPA2/WPA3) happens after association via EAPOL or SAE.

Step 3 – Association:

- The client sends an Association Request frame containing: supported rates, QoS capabilities, HT/VHT/HE capabilities, and RSN Information Element (security parameters).
- The AP evaluates the request and responds with an Association Response containing: Association ID (AID), supported rates confirmed, and status code (0 = success).
- Once associated, the client has an AID and is part of the BSS.

Step 4 – Security Authentication (EAPOL 4-Way Handshake for WPA2):

- After association, if WPA2-Personal is used, the EAPOL 4-way handshake begins to derive and install encryption keys.
- For WPA2-Enterprise (802.1X), EAP authentication with a RADIUS server happens first, followed by the 4-way handshake.
- For WPA3, SAE (Simultaneous Authentication of Equals) replaces the pre-shared key exchange.

Step 5 – DHCP and Network Access:

- Once keys are installed, the client sends a DHCP Discover to obtain an IP address.
- After receiving an IP, the client can communicate on the network.
- The client is now fully connected and data transfer begins.

6. Explain each step involved in EAPOL 4-way handshake and the purpose of each key derived from the process.

Answer:

The EAPOL (Extensible Authentication Protocol Over LAN) 4-Way Handshake is performed after 802.11 association to mutually verify knowledge of the PMK (Pairwise Master Key) and derive session encryption keys. It is used in WPA2-Personal and WPA2-Enterprise.

Key Hierarchy (Before the Handshake):

- **PMK (Pairwise Master Key):** In WPA2-Personal, $PMK = PBKDF2(PSK, SSID, 4096 \text{ iterations})$. In WPA2-Enterprise, PMK is derived from EAP authentication with RADIUS server. The PMK is 256 bits and is the root of all session keys.
- **PTK (Pairwise Transient Key):** Derived during the handshake. Used for unicast frame encryption between AP and client.
- **GTK (Group Temporal Key):** Used to encrypt broadcast and multicast frames — shared among all clients on the BSS.

The 4-Way Handshake Steps:

Message 1 – AP → Client (ANonce):

- The AP sends its ANonce (AP Nonce — a random 256-bit number) to the client.
- Purpose: Provides the client with the AP's random value needed to derive the PTK.
- At this point, the client now has: PMK + ANonce + SNonce (its own) + AP MAC + Client MAC → derives the PTK.
- PTK derivation: $PTK = PRF(PMK, ANonce, SNonce, AP_MAC, Client_MAC)$

Message 2 – Client → AP (SNonce + MIC):

- The client sends its SNonce (Station Nonce) to the AP, along with a MIC (Message Integrity Code) computed using the KCK portion of the derived PTK.
- Purpose: Proves to the AP that the client has the correct PMK (it could only compute a valid MIC if it derived the correct PTK).
- The AP now also derives the PTK using the received SNonce.
- The AP verifies the MIC — if valid, the PMK is confirmed to match on both sides.

Message 3 – AP → Client (GTK + Install PTK):

- The AP sends the GTK (Group Temporal Key) encrypted with the PTK's KEK (Key Encryption Key).
- The AP also sets the Install flag, instructing the client to install the PTK.
- A MIC is included to prove integrity.
- Purpose: Delivers the group key securely and signals the client to start using the PTK for encryption.

Message 4 – Client → AP (Confirmation):

- The client sends a confirmation ACK to the AP with a MIC.
- Purpose: Confirms that the client has received the GTK and installed the keys.
- After this, both AP and client have installed the PTK and GTK — encrypted communication begins.

Keys Derived and Their Purpose:

Key	Derived From	Size	Purpose
PMK	PSK + SSID (Personal) or EAP (Enterprise)	256 bits	Master key — root of all session keys; never transmitted over the air
PTK	$PRF(PMK + ANonce + SNonce + MACs)$	384–512 bits	Parent of KCK, KEK, and TK — used for unicast frame protection
KCK (Key Confirmation Key)	First 128 bits of PTK	128 bits	Computes the MIC in EAPOL messages 2, 3, 4 to verify integrity
KEK (Key Encryption Key)	Next 128 bits of PTK	128 bits	Encrypts the GTK in Message 3 so it is not sent in plaintext
TK (Temporal Key)	Remaining bits of PTK	128/256 bits	The actual encryption key for unicast data frames

Key	Derived From	Size	Purpose
			(AES-CCMP or AES-GCMP)
GTK (Group Temporal Key)	Generated by AP	128/256 bits	Encrypts broadcast and multicast frames sent to all clients in the BSS

7. Describe the power saving scheme in MAC layer and explore on the types of Power saving mechanisms.

Answer:

Wi-Fi radios consume significant power. The MAC layer includes power saving mechanisms that allow client devices to turn off their radio for periods of time, reducing battery consumption while maintaining network connectivity.

Basic Concept:

When a client enters power save mode, it notifies the AP by setting the Power Management bit in the Frame Control field of its transmitted frames. The AP then buffers any frames destined for that client while its radio is off.

1. Legacy Power Save (PS-Poll) – 802.11 Original:

- The client sleeps between beacon intervals (default 102.4 ms).
- It wakes up for every beacon and checks the TIM (Traffic Indication Map) — a bitmap in the beacon that indicates which clients have buffered frames.
- If the TIM indicates buffered frames for it, the client sends a PS-Poll frame to the AP to retrieve one frame at a time.
- Limitation: One PS-Poll per frame — high overhead for multiple buffered frames. High latency.

2. U-APSD (Unscheduled Automatic Power Save Delivery) – 802.11e WMM-PS:

- The client can trigger the AP to deliver all buffered frames at once by sending a trigger frame (a QoS data or null frame).
- The AP delivers all buffered frames for that AC (Access Category) in a burst, then sends an EOSP (End of Service Period) frame.
- Much more efficient than PS-Poll — reduces the number of PS-Poll exchanges.
- Used widely in voice and video applications (VoIP, video calls) since the client can trigger delivery at predictable intervals.

3. SMPS – Spatial Multiplexing Power Save (802.11n):

- Allows MIMO devices to turn off some of their antenna chains when not needed.
- Static SMPS: Client sends an SMPS Action frame to tell AP it is using single-stream mode — AP transmits single-stream to it.
- Dynamic SMPS: Client switches dynamically between single and multi-stream based on activity.
- Saves power when full MIMO throughput is not needed.

4. TWT – Target Wake Time (802.11ax / Wi-Fi 6):

- The most advanced power saving mechanism. The AP and client negotiate a specific schedule (TWT agreement) defining exactly when the client will wake up to send or receive data.
- Individual TWT: One-to-one agreement between AP and client — client wakes only at its scheduled TWT.
- Broadcast TWT: AP schedules a group of clients to wake at the same time.
- The client's radio can stay off for much longer periods between TWT sessions — dramatically extends battery life.

- Especially beneficial for IoT devices that only need to transmit small amounts of data at regular intervals.
- Can achieve up to 7× improvement in battery life compared to legacy power save.

8. Describe the Medium Access Control methodologies.

Answer:

Medium Access Control (MAC) methodologies define how devices share the wireless medium to avoid collisions and ensure fair access. Unlike wired Ethernet (CSMA/CD), wireless networks cannot detect collisions during transmission, so they use collision avoidance instead.

1. CSMA/CA – Carrier Sense Multiple Access with Collision Avoidance:

This is the primary MAC methodology in 802.11. Before transmitting, a device:

- Listens to the medium (carrier sense) — if busy, waits.
- Waits for DIFS (DCF Interframe Space) after the medium becomes idle.
- Selects a random backoff value from a contention window and counts down.
- Transmits only when the backoff counter reaches zero.
- The receiver sends an ACK after successful receipt. If no ACK is received, the sender retransmits with a larger contention window (binary exponential backoff).

2. DCF – Distributed Coordination Function:

- The foundational MAC method — built on top of CSMA/CA.
- All devices compete equally for the medium using random backoff.
- Used for best-effort data traffic.
- No central coordination — fully distributed.

3. PCF – Point Coordination Function:

- Optional, centralized MAC method where the AP acts as a Point Coordinator.
- Uses a Contention-Free Period (CFP) where the AP polls clients one by one.
- Eliminates contention during CFP — deterministic access.
- Rarely implemented in practice — superseded by HCF.

4. HCF – Hybrid Coordination Function (802.11e):

HCF combines contention-based and contention-free access for QoS:

- EDCA (Enhanced Distributed Channel Access): Extends DCF with 4 Access Categories (AC_VO, AC_VI, AC_BE, AC_BK). Higher-priority ACs use shorter AIFS (Arbitration IFS) and smaller contention windows, giving them statistical priority.
- HCCA (HCF Controlled Channel Access): Centralized polling by AP for guaranteed service periods. Used for deterministic QoS. Rarely deployed.

5. Virtual Carrier Sense – NAV (Network Allocation Vector):

- Devices that hear a frame update their NAV timer based on the Duration field in the MAC header.
- They defer transmission until the NAV expires, even if they cannot hear the actual transmission.
- Used with RTS/CTS to solve the hidden node problem.

6. OFDMA-based Access (802.11ax):

- Wi-Fi 6 introduced trigger-based uplink access via OFDMA.
- The AP sends a Trigger Frame that schedules multiple clients to transmit simultaneously on different Resource Units (RUs).
- This is a scheduled, non-contention-based method for uplink — dramatically reduces collisions in dense environments.

9. Brief about the Block ACK mechanism and its advantages.

Answer:

The Block ACK (Block Acknowledgment) mechanism, introduced in 802.11e and extended in 802.11n, allows a receiver to acknowledge multiple frames with a single ACK frame instead of sending an individual ACK after every frame. It works in conjunction with frame aggregation (A-MPDU).

How Block ACK Works:

Setup Phase:

- The sender initiates a Block ACK session by sending an ADDBA (Add Block ACK) Request to the receiver.
- The request specifies: Buffer Size (max frames in a burst, e.g., 64), Starting Sequence Number, and TID (Traffic Identifier).
- The receiver responds with an ADDBA Response accepting or rejecting the request.

Transmission Phase:

- The sender transmits a burst of frames (up to the negotiated window size) without waiting for individual ACKs.
- Each frame has a sequence number.
- When done, the sender sends a BAR (Block ACK Request) frame specifying the starting sequence number.

Acknowledgment Phase:

- The receiver responds with a Block ACK (BA) frame containing a bitmap.
- Each bit in the bitmap corresponds to a frame sequence number — 1 = received, 0 = missing.
- The sender retransmits only the frames marked as missing (0 bits).
- The teardown is done via DELBA (Delete Block ACK) frames.

Types of Block ACK:

- Immediate Block ACK: BA is sent immediately after the BAR.
- Delayed Block ACK: BA may be sent after a delay (rarely used).
- Compressed Block ACK: Uses a 64-bit bitmap — acknowledges up to 64 frames (802.11n).
- Multi-TID Block ACK: Acknowledges frames across multiple Traffic Identifiers in one BA (802.11n).
- HE Block ACK (802.11ax): Extended bitmap for larger aggregation windows.

Advantages of Block ACK:

- Reduced Overhead: One Block ACK replaces up to 64 individual ACK frames — dramatically reduces the number of control frames on the medium.
- Higher Throughput: Less time spent on ACK transmissions = more time for data.
- Efficient with Aggregation: Block ACK is designed to work with A-MPDU — together they are the primary reason 802.11n/ac/ax can achieve hundreds of Mbps.
- Selective Retransmission: Only missing frames are retransmitted, not the entire burst.
- Lower Latency: Fewer round-trips between sender and receiver for acknowledgments.

10. Explain about A-MSDU, A-MPDU and A-MSDU in A-MPDU.

Answer:

Frame aggregation is a key 802.11n/ac/ax optimization that combines multiple frames into a single transmission, reducing the overhead from inter-frame spaces, headers, and ACKs. There are two types of aggregation and a combined form.

1. A-MSDU – Aggregate MAC Service Data Unit:

- Aggregates multiple MSDUs (the data payloads from upper layers, e.g., IP packets) into a single MPDU (one MAC frame).
- Structure: Single MAC Header → [MSDU 1 | MSDU 2 | MSDU 3 | ...] → Single FCS.
- Each subframe has its own MSDU header (Destination Address, Source Address, Length).
- Maximum size: 7935 bytes (802.11n), 11,426 bytes (802.11ac/ax).
- All MSDUs must be for the same receiver and same TID (Traffic Identifier).
- Advantage: Reduces MAC header overhead — one MAC header for multiple payloads.
- Disadvantage: If any part of the A-MSDU is corrupted, the entire A-MSDU must be retransmitted (single FCS covers everything).
- No encryption boundary per MSDU — the entire A-MSDU is encrypted as one unit.

2. A-MPDU – Aggregate MAC Protocol Data Unit:

- Aggregates multiple MPDUs (complete MAC frames, each with its own header and FCS) into a single PHY layer transmission (PPDU).
- Structure: PLCP Header → [MPDU Delimiter | MPDU 1 | MPDU Delimiter | MPDU 2 | ...] → End.
- Each MPDU has its own MAC header, FCS, and is independently encrypted.
- Maximum number of MPDUs: 64 (802.11n), 256 (802.11ac/ax).
- Maximum A-MPDU size: 65,535 bytes (802.11n), 1,048,575 bytes (802.11ac).
- Works with Block ACK — the receiver can acknowledge which individual MPDUs were received using the bitmap.
- Advantage: If one MPDU is corrupted, only that MPDU needs retransmission — not the entire aggregate.
- More robust than A-MSDU for error-prone environments.

3. A-MSDU in A-MPDU (Two-Level Aggregation):

- This is the combination of both aggregation methods for maximum efficiency.
- Each MPDU in the A-MPDU can itself contain an A-MSDU — so multiple MSDUs are packed into each MPDU, and multiple such MPDUs are packed into one A-MPDU transmission.
- Structure: A-MPDU { [MPDU with A-MSDU 1] | [MPDU with A-MSDU 2] | ... }
- This provides: Reduced MAC header overhead (from A-MSDU) + Independent per-MPDU error recovery (from A-MPDU) + Maximum data density per PPDU.
- Used in 802.11n/ac/ax for maximum throughput scenarios.
- Limitation: Only applicable to frames that are not management frames and must share the same receiver and TID within each A-MSDU.

Feature	A-MSDU	A-MPDU	A-MSDU in A-MPDU
What is aggregated	Multiple MSDUs into one MPDU	Multiple MPDUs into one PPDU	MSDUs within MPDUs within one PPDU
MAC headers	One for all	One per MPDU	One per MPDU (MSDUs share it)
FCS	One for all	One per MPDU	One per MPDU
Error recovery	Entire frame retransmitted	Per-MPDU retransmission	Per-MPDU retransmission
Max size	7935–11,426 bytes	Up to 1 MB (802.11ac)	Maximum possible
Block ACK compatible	No (single ACK)	Yes	Yes